A Library-Free Chemical Elucidation Probe and Its Unclosed Enabling Gap

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Abstract

The requested result is a new means of discovery or production that opens consequential work beyond competent existing methods, followed by accepted delivery, complete-cost surplus and earned-funded successor construction. This campaign has not achieved it. After early conventional routes removed two software gaps, the investigation moved to chemical identification outside a reference library. Real source-bound mass spectra were acquired and a runnable conventional baseline was executed before claiming invention. An initial 27-record, three-identity collision-energy study was already resolved by ordinary entropy matching at one recorded energy. The library-free probe then combined MAYGEN constitutional-graph enumeration with local MetFrag fragmentation scoring on two source-verified anonymous queries of formula C7H17N. All 89 generated graphs were independently verified and scored, but the truths remained in top-score tie sets of eight and twelve at every tested depth. The registered primary criterion was internally impossible and is adjudicated UNKNOWN_INVALID_CRITERION. No new enabling construction or independently useful chemical delivery is qualified at this checkpoint.

The work preserves a distinct reusable effect-factorization compiler, its 90-check finite verification, a real streaming adapter and technical transfer, strictly as supporting engineering. The accepted compatibility report entered a 15-check simulated exchange; its canonical JSON content matched the accepted report, but raw byte identity did not, and the exact-byte claim remains unqualified. No buyer cohort, supported price, complete-cost crossing or qualified simulated earnings was established. The prior release's small identity gap was separately closed without rewriting the original overrun. This paper records the executed scientific probe, supporting constructions, unresolved obligations and bounded continuation; publication is not completion of the invention mission.

1. Outcome, authority and inherited failures

An invention flywheel requires an evidenced beneficiary outcome, an enabling instrument, independent acceptance, rational buyer and operator participation, complete-cost surplus, unrestricted earned funds and additional construction that benefits fresh work. These links must concern the same useful capability. Results from unrelated demonstrations cannot be joined into a fictitious commercial achievement.

The starting revision was cd90c81ab757a7042bf4d0384e8e1a7e6c2a9557, the merge of Lab PR #10. Strong ridge controls defeated that campaign's primary predictor criterion, ordinary reuse tied inheritance, and neither a commercial cohort nor earned-funded compounding was established. Its developer-supplied group averaging was a conventional symmetry repair. All original sources, failures and releases remain intact. No new predictor development or confirmation was conducted; any required historical CI replay retains its exposed reproduction status.

The 30 October instruction authorized public implementation, bounded experiments, ordinary reviewed publication and simulated commerce. It excluded new spending, paid provider calls, customer outreach, real transfers, credential grants, broader access, deletion of unique evidence and weakened safeguards. The fresh allocation began at 04:40 UTC: 90 wall minutes, a 12,000 CPU-second capacity, 4 GiB additional storage and

an 8 GiB network-receive limit. Development stops at 05:40 UTC, reserving 30 minutes for reproduction and publication before 06:10 UTC. These are limits, not measured costs. Earlier read-only inspection and a separately authorized prior identity audit are reported separately.

One lead connected opportunity research, construction, verification, economics and release work. The roles shared a model family, libraries, host and filesystem. Verification is procedurally separate, not external certification or an operating-system-enforced information barrier. Exposed cases remain exposed across subsequent runs.

2. Screen, falsification and the consequential redirection

Three materially different possibilities were screened before executable commitment: physical lead service-line verification, accessible PDF delivery, and behavior-preserving dependency migration. EPA and Philadelphia sources establish physical-observation obligations for hidden pipes; W3C and Canadian guidance establish semantic and assistive-technology obligations for PDF delivery. Neither complete field access nor complete assistive-technology acceptance was available. Those possibilities remained unadmitted rather than receiving simulated substitutes counted as real delivery.

The cheapest executable dependency probe reproduced a break after an httpx-only bump. The original pinned environment passed nine upstream documentation tests; the direct bump failed during test collection before test assertions executed. An ordinary coordinated FastAPI/Starlette upgrade passed all nine without changing them. A separate frozen verifier then passed 18 upstream tests and 34 additional behavior checks across original and upgraded environments, with zero skips. Thus the complete conventional route removed the proposed gap. Compatibility does not qualify vulnerability remediation or production support.

A streaming-lifetime probe likewise found that ordinary explicit commit plus request-scoped cleanup satisfied both durability-before-response and resource-live-through-stream obligations. These results are preserved as supporting machinery, not relabeled a consequential flagship.

The investigation was redirected to analytical teams facing unknown reaction products, impurities or natural products outside spectral libraries. The beneficiary outcome is a defensible structural assignment from admissible measurements, with unresolved alternatives and downstream responsibilities stated. Candidate enumeration, fragment ranking and a source-authenticated peak list are useful components; none alone is that complete delivery.

3. Real-data access and the library-matching falsifier

Source and cohort admission

The initial scientific probe used MassBank data pinned at d4cccb5b6756bb14d52568c82720dde1dd3d42e1. Metadata selection preceded intensity matching. It fixed full InChIKey identity, formula, ion mode, adduct, precursor, exact instrument string and fragmentation field. The selected C19H29NO2 group contains three full identities at nine common exact nominal-percent collision-energy settings, all measured on the reported Exploris 240 Orbitrap under positive [M+H]+ HCD conditions. All 27 records were retained.

The separate verifier checked their exact Git objects, all peak values, metadata, coverage, authorship and licenses. The observations JSON is bound to SHA-256 dc2b68d35fec47c16c4a5b2805cd6f1d8c415a91f109 5d893d6abed1e771a7a1. Reported standard-compound confidence is source attribution, not new chemical authentication.

Every identity-energy cell has one record. Distinct accessions, dates or peak hashes do not establish independent injections. There is no independent query replicate, and all 27 records are explicitly development data. Nominal percentages were not globally converted into eV; scaled, ramped and other energy protocols were not silently equated.

Strong ordinary baseline

The baseline executed maintained ms-entropy 1.5.3 with 0.02 Da matching tolerance and default cleaning. The verifier independently parsed raw records and reproduced all nine similarity matrices to within 1e-10 using the same disclosed library. An ordinary single higher-energy record already separates the three exact reference profiles. This falsifies the noiseless closed-library enabling-gap hypothesis.

Self-matching is not unknown-sample accuracy. Threshold sensitivities from 0.90 to 0.99 are numerical probes, not calibrated noise or identification guarantees. Replaying a recorded energy is simulated acquisition, not a new physical experiment; an energy-setting count is not an established injection count, scan count or cost. The published longitudinal-fragment-profile method was inspected as prior art, but its unavailable complete plasma pipeline was not claimed as an executed comparator.

4. Library-free constitutional elucidation: the executed baseline

Outcome and available information

The distinct question was whether candidate structures could be constructed without retrieving the desired answer from a reference library, then distinguished using the measured fragments. Two anonymous C7H17N queries each contain nine source-derived energy observations. The separate verifier matched all 18 records to the pinned source while initially holding back their full constitutional structures. Formula and measured peaks were admissible builder inputs; concealed labels were reserved for the subsequent truth audit.

This is a bounded constitutional-graph population. It does not establish stereochemical elucidation, mixture deconvolution, structure authentication or general unknown-impurity identification. The two source-reported standards are not fresh physical acquisitions or a statistically representative industrial cohort.

Executed conventional construction path

The ordinary baseline uses MAYGEN 1.8 to enumerate formula-compatible constitutional graphs into SDF, then MetFrag 2.6.11 against that local generated file. Candidate generation uses no queried structure database. This is a competent formula-to-fragment route, not a qualification of the strongest complete physical elucidation service. Unexecuted current commercial or AI-native routes are not presumed to fail. The tool binaries, source URLs, versions and licenses are pinned; MAYGEN is MIT-licensed and the MetFrag release is LGPL-2.1-or-later.

For each query, positive intensities are normalized within energy observations, pooled and clustered within 0.001 Da by the registered implementation. MetFrag receives the neutral formula and monoisotopic mass, local SDF, 0.001 Da absolute and 5 ppm relative fragment tolerances, positive ion mode and FragmenterScore. Depths 2, 3 and 7 are separately identified development settings; no successful depth is silently substituted for failed runs.

Each process is bounded to two active processors, 1 GiB Java heap, 90 CPU seconds and 90 wall seconds. Local fragmentation is configured without a usable network proxy. The process records stdout, stderr, settings, peaks, SDF, source identities, exit state and measured resources. Tool setup and complete monetary cost remain separate.

Executed result and interpretation

MAYGEN generated 89 constitutional graphs. An independent Python Prufer/tree enumeration, separate from MAYGEN/CDK, produced the same complete graph universe. All 89 graphs were scored for each query and both held-back constitutional truths were included. The three declared MetFrag depths produced the same observed ambiguity:

MetFrag depth	q01 true rank interval / top-score tie	q02 true rank interval / top-score tie	Complete enumeration and scoring
2, 3 and 7	1-8 / 8 graphs	1-12 / 12 graphs	89/89 for each query

Both truths receive score 1.0, but neither is uniquely identified. The top-score tie sets are descriptive results, not calibrated coverage or posterior probabilities. This establishes residual ambiguity for the tested local fragmentation score, not failure of all conventional chemistry or proof that a new measurement instrument works.

The original registered primary endpoint was internally incompatible. It required a tie-inclusive first-ten-rank shortlist and at least 90% reduction from 89 candidates; the first condition retains at least ten candidates, while the second requires at most eight. Literal false fields in the original reports remain preserved but cannot serve as a scientific falsification. The adjudicated primary decision is UNKNOWN_INVALID_CRITERION. The valid descriptive counts above are kept separate; no favorable replacement endpoint or new confirmation budget is introduced after exposure. The verifier then constructed an exact rational-arithmetic preflight guard, with six passing tests. Running it on the unchanged original contract refuses the incompatible 10-versus-8 counts. A prospective first-rank proposal is only arithmetically possible, not guaranteed under ties, and remains unrun on fresh evidence. This repairs a research-process guard; it is developer/verifier-authored and establishes no autonomous policy advantage or chemical enabling capability.

The initial rank parser also refused explicit hydrogen atoms in the MetFrag SMILES before interpreting Score fields or computing truth ranks. Its replacement validates hydrogen degree and heavy-atom valence, removes explicit hydrogens for constitutional canonicalization, and verifies all 89 graphs against the independently checked SDF universe before rank interpretation. Original refusals and repaired parser identities remain preserved. Source labels and all truth outcomes are now exposed; any subsequent adaptation uses development evidence.

The current answer to whether a new consequential enabling construction is qualified is **no**. A conventional failure can specify a missing instrument, but does not demonstrate that a proposed repair works. A conventional success removes the claimed gap. Neither outcome licenses a profitable business or beyond-human-foresight claim.

5. Construction retained as supporting engineering

The separate lifetime instrument takes typed acquire, prepare, durable, consume and release effects, coupled blocks and explicit factorization permissions. It computes role mappings, searches block arrangements, separates authorized compound effects, and generates executable plans and a reusable operator. A real SQLite/ASGI adapter executes the generated before-consume plan, streams from a live resource and cleans up at request exit. Four development cases pass: success, actual deferred commit failure, midstream failure and injected midstream interruption. Partial output is never accepted as full delivery.

Operation semantics, donor roles, constraints, grammar and search code were developer supplied. Misleading names do not drive correspondence. This is bounded executable adaptation, not an autonomous discovery of a new mathematical principle. Plans execute through at most five fixed capabilities and 64 bounded string values; no generated Python, shell, network access, dynamic imports or caller-selected paths are allowed.

Criticism reproduced a qualification defect: initial consumption faults occurred before any emitted record, so they did not test partial-delivery cleanup. The typed repair obligation required faults after a visible chunk, preservation of early-failure tests, exactly-once release and truthful partial-output status. Original source, reproduced traces, repaired source and results are retained. Injected exceptions are not actual process cancellation.

The frozen separate temporal verifier passed 90 finite checks, including 13 real executions, handle instrumentation, retained SQLite/file bytes, malformed specifications, stale or forged plans, corruption and applicability failures. Its panel remapped an in-memory generated operator; that alone is not persistent memory retrieval. The separate transfer runner performs actual retained-file acquisition, retrieval, remapping, construction and execution on a file-domain successor.

The actual retained-file path acquired one source-bound record, retrieved it with no retrieval failures, remapped it to a fresh framed-file export and executed that construction. The frozen old toolbox could not construct a valid unsplit plan; inherited retrieval, equal-content ordinary reuse, reset with equal tools and explicit conventional split all accepted the same output. The complete transfer runner used 0.043 wall seconds and

0.019 process CPU seconds. This tiny authored file task is a technical transfer check, not a consequential chemistry successor or earned-funded construction.

The explicit conventional split meets the same contract, and equivalent-content ordinary reuse ties the remapped content. A successful transfer is reusable content; it does not establish competitive or after-cost experience benefit. No learned discovery-policy improvement is credited to developer-authored search code. Technical successor work uses the research allocation, not qualified commercial earnings.

6. Independent demand, complete obligations and economics

Analytical elucidation is consequential and commercially offered. Northwestern IMSERC describes small-molecule identification using MS and NMR. Oregon State offers full elucidation with instrument and staff charges; its concentrated example acquires six standard spectra in under one hour. NorrChemica lists a conditional EUR 599 HRMS/proton/carbon-NMR assignment package and EUR 150 rest-of-world order fee; full elucidation requiring additional two-dimensional experiments remains quote-only. These are distinct obligations, not a universal EUR 749 unknown-structure guarantee. Vendor testing ranges with unspecified currency are not converted into precise prices.

A complete strong route may require isolation, physical measurements, suitable standards, interpretation, competing-structure checks, uncertainty and a report. An ordinary LC-MS/MS sample tariff cannot price that entire service. ICH/FDA impurity guidance establishes product-specific identification and qualification responsibilities; it grants no regulatory or manufacturing-release qualification to this software.

No frozen loss-making customer cohort, committed buyer, realized purchase or supported AGI ALPHA price exists. Value created for a chemist, benefit to a buyer and revenue captured by this enterprise remain separate. Free tools, competent physical/software combinations and non-entry remain admissible alternatives.

For price P, paying count q, refund fraction r, recorded cost lower bound L and remaining complete cost K, a necessary profit condition is:

$$q(1-r)P - L - K > 0.$$

The evidence provides no finite upper bound for K. It includes inference, failed search, acquisition, construction, verification, integration, deployment, storage and bandwidth, maintenance, support, disputes, fees, financing and terminal obligations. A missing bill is not zero. Participation must also beat alternatives after trust discounts, switching, procurement and risk. Negative-to-positive complete-cost crossing, investment attractiveness and value capture therefore remain UNKNOWN.

The supporting compatibility offering was separately bound to its limited technical receipt and passed 15 simulated exchange checks. All six fixture inputs remain in the record. Four arbitrary one-dollar simulated releases collected four dollars before correlated reversals; final collections and unrestricted earnings were zero. No unknown lifecycle expenses were posted, so intermediate ledger profit was partial fixture accounting. Strict post-reserve earned control was zero even before reversal.

A separate criticism establishes a delivery-identity defect: the accepted report had 11,303 raw bytes, while the released canonical JSON had 6,008 bytes. Parsed content matched but raw hashes did not. The 15 checks qualify declared canonical-content and settlement behavior; the exact-byte claim is false and remains unqualified. Original source and receipts are preserved. The scientific query received no buyer acceptance or earnings from this unrelated compatibility exercise.

Qualified simulated earnings are zero. Opening capital, custody, borrowing and reserve releases cannot fund a claimed commercial successor. Earned-funded capability construction is UNRUN. No real outreach, order, payment or live commercial success occurred.

7. Verification, resources and publication identity

Scientific source checks use separate parsers and original Git objects. The ordinary entropy replay shares the underlying algorithm/library, and all roles share infrastructure. Procedural separation is disclosed.

Source-derived labels are not independent chemical authentication, and replaying exposed measurements is not protected physical confirmation. Statistical alpha spent is zero; no population-superiority claim or reset of historical error is asserted.

The prior four-CPU-second audit remains historically unqualified. A separate new 120-CPU/150-wall identity audit verified thirteen published assets (216,893,436 bytes), all 6,364 archive payloads and 6,363 frozen sources against external digests. It used 5.605 measured CPU seconds including download and supervision. Original failures and the old release are unchanged; no predictor ran. The source-bound receipt and full accounting remain in the evidence manifest.

Receipts record prospective bounds, conservative reservations, wall/CPU usage and output bytes, including failed setups and incomplete commands. Per-process limits do not establish full-host or descendant accounting. Initial setup, actor inference, infrastructure and CI remain incompletely priced; missing costs are not zero.

The final local connected reproduction passed against the externally pinned 92-file freeze: 17 unit tests, 90 exposed lifetime checks, six ordinary and four generated ASGI cases, actual and independently replayed spectra, fresh original/upgraded compatibility environments with 18 upstream tests and 34 behavior checks, conditional exchange, and all six Java query/depth outputs. The driver recorded 39.801 wall seconds; its separately timed chemistry subprocess consumed 56.317 waited-child CPU seconds across bounded parallel work, and parent self CPU was 0.358 seconds. These quantities must not be added to enclosing totals again. The pre-publication workspace snapshot was 1,437,518,064 bytes including inherited source and reproducible tools; full monetary and host-wide costs remain unknown.

Publication review reproduced an early wrapper import and an archive reread race. New versions verify source before importing the wrapper and digest/extract one captured archive snapshot with exact payload coverage. Original defects remain preserved. A repaired identity-layer probe passes but stubs runtime; it does not replace the full reproduction. Hosted CI and actual release-artifact qualification remain separately pending at manuscript freeze.

The LaTeX, PDF, browser HTML and native GitHub Markdown editions are bound to the same evidence manifest. Mechanical binding and visual QA do not certify the truth of scientific or economic claims. Exact-head tests, ordinary review, normal merge and published-package extraction/reproduction remain separate qualification steps; historical assets are not rewritten.

8. Decisions and bounded continuation

Required decision	Status at this checkpoint
Engineering qualification	Scoped checks pass; package separate.
Consequential enabling capability	Not established; no qualified new means.
Combination benefit	Conventional combination ties.
Experience benefit	Ordinary reuse and reset tie.
Conditional commercial viability	Unknown; no admitted cohort or complete costs.
Investment attractiveness and AGI ALPHA capture	Unknown; alternatives remain admissible.
Qualified simulated unrestricted earnings	Zero.
Earned-funded successor benefit	Unrun; no qualified earnings.
Live commercial success	Unrun; no real orders or transfers.

Install the pinned free dependencies and execute into new paths:

```
python -m pip install -r requirements-cycle007.txt
python run_cycle007.py --output /tmp/alpha007-reproduction-NEW \
  --expected-freeze-sha256 \
  0001664b875aec9854e6cd033a97b28d80bc47bfa9a515b63c037db7e0598834 \
  --with-compatibility --chemistry-tools-dir /tmp/alpha007-tools-NEW \
  --download-tools
```

Obtain the expected digest through the trusted versioned GitHub paper or release channel separately from the archive being checked. Copying a hash from an untrusted archive does not establish external authority. The command verifies that supplied identity before candidate imports and rechecks source identities at completion. It reproduces exposed cases; it does not create new scientific confirmation or replenish the error budget.

The next action must follow the measured de novo baseline, not another architecture. If ambiguity survives its strongest complete route, freeze the exact missing observation or constructive defect, build the smallest instrument that could remove it, and test its effect on independent acquisitions under the same obligations. If the conventional route meets the objective, retain it and do not force flagship status. New resource or access needs must be stated precisely; this checkpoint does not imply a background continuation.

References and evidence map

All public-source observations are dated 30 October 2026 unless explicitly historical. Original authors, record licenses and exact tool identities remain with the source data. The final input manifest provides file-level hashes and a stable source anchor.

- MassBank, source data; pinned source revision and original records are reproduced with attribution.
- MAYGEN, constitutional structure generator; MetFrag, local fragmentation and ranking.
- Longitudinal MS/MS prior art; unavailable full-pipeline execution is disclosed.
- Northwestern structure elucidation; Oregon State small-molecule characterization; NorrChemica current service scope.
- FDA, Q3A impurities guidance.
- FastAPI, dependency lifetime semantics; its documentation and tests are supporting comparator evidence.
- Source-bound records: opportunity screen, MassBank admission/source/baseline reports, de novo input/tool/enumeration/truth receipts, temporal and compatibility reviews, economic identity adjudication, all original failures, resource ledger and reproduction receipt.

Stable experimental source anchor: 15ee8b0fed9e50db4c3e4a3680bfc00c78287307.

Recorded allocation and authority; Opportunity screen; MassBank source qualification; Independent ordinary spectral baseline; Library-free depth-seven truth audit.

Invalid-primary-criterion adjudication; Preserved canonical-byte identity defect; Complete local connected reproduction; Explicit campaign decisions; Prospective costs and demand limitations.